

Binomial Coefficients in Probability and Expansion

Pascal's triangle, binomial expansions, single terms, and binomial probability

Directions:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the ways to choose k objects from n .
- Give exact values for counts and expansions; round probabilities as each problem asks.
- The four subskills are mixed on purpose after problems 1–2 — decide which tool each problem needs before you start computing.

1. Evaluate $\binom{6}{2}$.

Answer: _____

2. Expand $(x + 2)^3$ and write the result in descending powers of x .

Answer: _____

3. A fair coin is flipped 5 times, giving 32 equally likely sequences. Exactly $\binom{5}{2} = 10$ of them contain exactly two heads. Find the probability of getting exactly two heads.

Answer: _____

4. In the expansion of $(x + 3)^5$, find the coefficient of x^3 .

Answer: _____

5. Row 7 of Pascal's triangle contains the entries $\binom{7}{2} = 21$ and $\binom{7}{3} = 35$. Use the addition rule for Pascal's triangle to find $\binom{8}{3}$ without computing any factorials.

Answer: _____

6. Expand $(2x - 1)^4$ and write the result in descending powers of x .

Answer: _____

7. A basketball player makes each free throw with probability 0.4, independently. She takes 5 free throws. Find the probability that she makes exactly 3 of them. Round to four decimal places.

Answer: _____

8. In the expansion of $(x^2 + 3)^3$, find the coefficient of x^4 .

Answer: _____

- 9.** The same player takes 5 free throws, making each with probability 0.4.
- (a) Find the probability that she makes at least 4 of the 5. Round to three decimal places.

- (b) Explain why this answer needs two binomial terms added together rather than one.
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- 10.** Consider $(1 + x)^4$.
- (a) Expand it in ascending powers of x . _____
- (b) Explain what the coefficient of x^k counts, in terms of choosing from the four factors.

11. Consider row 6 of Pascal's triangle: 1, 6, 15, 20, 15, 6, 1.

- (a) Find the sum of the entries in that row. _____
- (b) Explain why the sum of row n is always 2^n .

12. Synthesis. Consider the expansion of $(2x + 3)^7$.

- (a) Find the coefficient of x^5 . _____
- (b) Explain why the power of 3 in that term must be 2.